



Grain boundary meshes in Kanapy

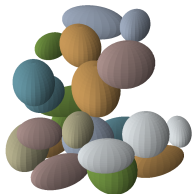
A proposal for local repair of point contacts

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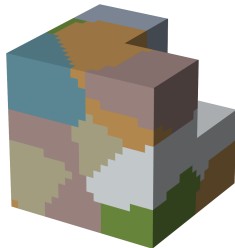
1 October 2026

The goal: shared grain surfaces suitable for volume meshing



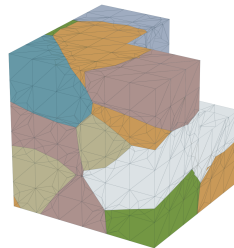
Packed ellipsoids

Positions and shape information



Voxel assignment

Grain labels on a regular grid



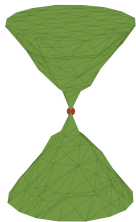
Shared APD surfaces

Triangles on a shared partition

Voxel labels and surfaces discretize the same APD. The goal is FE meshing from smooth interfaces.

Kanapy 6.5.5; one generated RVE, 23 grains, $24\ \mu\text{m}$ box. One octant is omitted in voxel and surface views. The tetrahedral background is a reconstruction scaffold, not an FE mesh.

The reported defect: one grain touches itself at a point



One grain, two lobes, one shared point

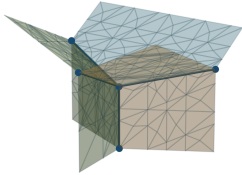
Two parts of one grain share only one voxel or tetrahedron node.

- The connection has no cross-sectional area.
- The grain boundary has a point self contact.
- A usable solid needs a finite connection or a different ownership assignment.

Target: repair the grain partition before final meshing.

Controlled APD generated with Kanapy from the existing analytical pinch regression. It illustrates the reported defect; it is not Hartmaier's unavailable failing RVE.

Normal grain junctions must remain allowed

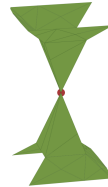


Different grains meet

Ordinary grain junctions

Check the boundary of *each grain*. The entire interface network contains legitimate triple lines and junctions.

Both geometries generated with Kanapy. Left: internal interfaces of four distinct grains, with box faces omitted; blue curves mark ordinary junctions.



One grain touches itself

Local boundary of the same grain is pinched

A vertex test detects what edge counts can miss

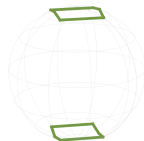


Regular boundary vertex

One connected cycle of incident triangles

- Require one connected cycle in each closed grain boundary vertex link.
- Also check volume connectivity through shared faces and geometric intersections.

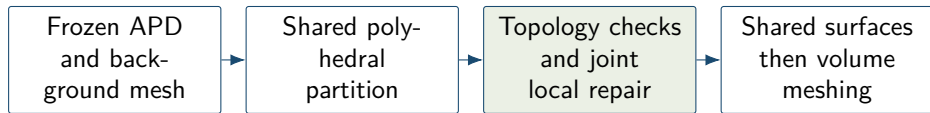
Links extracted from actual Kanapy triangles and normalized for display. Existing APD shell checks and surface validation check edge incidence; a complete vertex-link check is a proposed addition.



Point contact

Two separate cycles around the same vertex

Where the proposed repair fits into Kanapy



Already available

Work proposed here

APD costs, volume fitting

Freeze the source and report complete local topology

Shared polyhedral partition

Modify ownership in a bounded defect neighbourhood

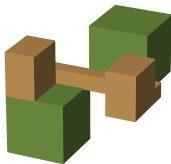
Lloyd conditioning

Compare it as a baseline; add explicit topology acceptance

Surface regularization and Gmsh

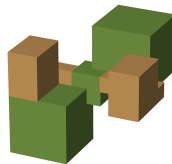
Recheck repaired geometry; evaluate FE volume meshing separately

Code: `power_diagram.py`, `apd_mesh.py`, `geometry_conditioning.py`, `surface_regularization.py`, `gmsh_remeshing.py`. Current Gmsh integration generates surface triangles.



Original partition

A has a point contact; **B** is connected



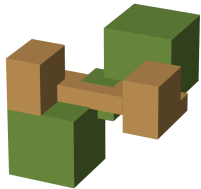
Candidate: widen only **A**

A is joined; **B** is disconnected

This candidate removes the point defect but must be rejected: grain **B** loses its connection.

Constructed 11^3 voxel partition rendered with Kanapy. Surrounding grain C is hidden in the views but included in every check.
Component counts use shared-face adjacency.

A joint candidate reconnects A and preserves B



Widen **A** and reroute **B**

Kanapy's exhaustive local voxel boundary test reports zero defect-incident voxels for all three grains. This is a discrete example, not a continuous APD repair or an FE result.

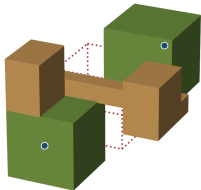
Measured on this constructed example

Grain	A	B	C
Face components	1	1	1
Defect voxels	0	0	0
Volume change	+4.69%	+4.44%	-0.69%

Ten of 1331 voxels change ownership. Every voxel still has exactly one grain label. **This**

demonstrates a joint edit. Automatic search and admissible volume and shape budgets remain to be developed.

Generate candidates inside a bounded neighbourhood



Dotted box: editable neighbourhood

Blue points: protected core locations

Proposal: use existing clipped APD polyhedra as initial cells. Conforming subdivision may be needed. The candidate search is bounded; it does not guarantee that an admissible repair exists.

1. Collect cells incident to the defect, plus a halo.
2. Freeze labels outside the patch and selected grain cores.
3. Try a finite-width neck and joint neighbour adjustments.
4. If appropriate, try reassigning a small expendable fragment.
5. Validate every affected grain. Refine or grow the patch within a fixed budget; otherwise report unresolved.

Choose the smallest admissible ownership change

Cell a has volume $|a|$ and label g :

$$z_{ag} \in \{0, 1\}, \quad \sum_g z_{ag} = 1.$$

$$V_g = \sum_a |a| z_{ag}.$$

Rank candidates by changed volume and disagreement with the frozen APD costs.

First implementation: deterministic

candidate generation with strict acceptance checks.

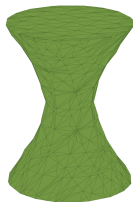
Ownership constraints prevent gaps and overlaps on a valid conforming cell complex. Manifoldness and connectivity require separate checks; this is not an ordinary unconstrained graph-cut problem.

Hard acceptance checks

- Valid boundary links and grain connectivity.
- Fixed exterior labels and protected cores.
- Cumulative volume bounds and consistent periodic copies.
- Required neck size and geometric modification bounds.

A low objective value cannot override these checks.

The difficult tradeoff: topology versus morphology



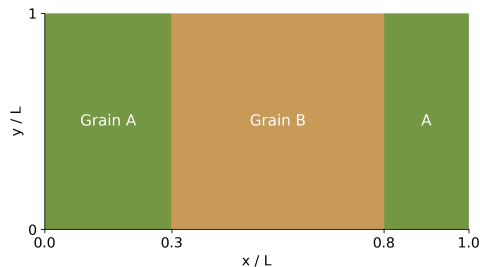
A finite neck changes the grain geometry

Requirement	What must be measured
Volume fidelity	Per-grain volume bounds accumulated from the frozen baseline
Shape fidelity	Centroid, covariance, aspect ratio and boundary displacement
Local structure	Neighbour changes, interface areas and junction motion
Resolvable neck	Cross-section relative to the intended FE mesh size

A small global volume change can still alter a local load path. Mechanical sensitivity must be tested.

The illustrated neck is an APD weight control and moves the whole interface. It is not a completed local repair. Tolerances must be chosen for the intended simulation.

Periodic grains must be checked across opposite box faces

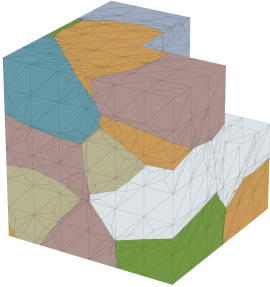


Kanapy periodic APD slice: the two green fragments
belong to the same grain

Constructed two-grain periodic APD evaluated directly by Kanapy. Start the repair prototype with interior, nonperiodic defects; periodic cases are a separate acceptance stage.

- Identify opposite faces before counting grain components or boundary links.
- Synchronize edits and shared geometry in all corresponding periodic copies.
- Do not treat a box cut as a physical grain boundary.

One repaired partition must feed every downstream representation



Shared triangles from the current
Kanapy geometry builder

Current Kanapy output is a surface mesh. Planned volume-meshing comparators: CGAL Mesh_3 and XtalMesh.

1. Store the repaired labels and cell complex as the authoritative geometry.
2. Extract each interface once from neighbouring cells with different labels.
3. Regularize and remesh surfaces, then repeat topology and intersection checks.
4. Evaluate a conforming volume mesher and check element quality, grain labels and periodic matching.

Reapplying the original APD minimum inside the repaired patch would undo the repair.

Test against known defects and complex RVEs

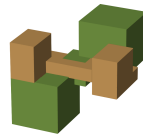
Cases	Comparisons	Recorded outcomes
Point and edge contacts; normal junction controls	Current pipeline; existing conditioning; local repair	Defects removed, new defects, unresolved cases
Fixed packed APDs at sev- eral resolutions	Freeze centres, matrices and weights; refine background	Approximation error separated from repair changes
Complex and periodic RVEs	Repeated seeds and paired geometry	Per-grain volume, moments, adja- cency and runtime
Meshed repaired partitions	Identical loading and mate- rial assignment	Positive Jacobians, mesh conver- gence, elastic response and local stress

First deliverable: a complete combinatorial report for the reconstructed partition and a bounded repair prototype. Validation on Hartmaier's case needs its saved input.

The benchmark will record unresolved cases, geometric changes, downstream mesh quality and mechanical sensitivity.

1. Complete the per-grain topology report and save failing inputs.
2. Implement the smallest interior repair patch with full neighbour checks.
3. Add volume and morphology budgets, refinement and rejection logic.
4. Extend to periodic defects; rebuild shared surfaces.
5. Benchmark volume meshing and mechanical sensitivity.

Each stage should return a saved geometry and a report that can be inspected.



Constructed joint edit

Decisions to agree:

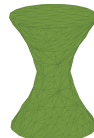
allowable volume and shape changes, treatment of small fragments, and target FE mesh scale.



$\Delta w < 0$: separated lobes



$\Delta w = 0$: point contact



$\Delta w > 0$: finite neck

$$q_i(\mathbf{x}) = (\mathbf{x} - \mathbf{c}_i)^T A_i (\mathbf{x} - \mathbf{c}_i) - w_i, \quad g(\mathbf{x}) = \arg \min_i q_i(\mathbf{x}).$$

With coincident centres $(.5, .5, .5)$, $A_1 = I$, $A_2 = \text{diag}(2, 2, .5)$:

$$\overline{\Omega_2} = \{\mathbf{x} : (x - .5)^2 + (y - .5)^2 - \frac{1}{2}(z - .5)^2 \leq \Delta w\}.$$

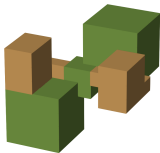
Existing regression: `test_power_diagram_diagnostics.py`. Surface geometry uses the grain closure; pointwise ties go to the first grain. These deliberately coincident seeds are a control, not packed particles.

For atomic cells a in the patch, propose:

$$E(z) = \alpha \sum_a |a| [g_a \neq g_a^0] + \beta \sum_a \int_a (\tilde{q}_{g_a}(\mathbf{x}) - \min_j \tilde{q}_j(\mathbf{x})) d\mathbf{x}.$$

- Freeze piecewise-affine costs $\tilde{q}$ and retain original clipped-atom boundaries during refinement.
- Integrate affine cost differences using atom volumes and centroids; normalize objective terms.
- Enforce volume bands $|V_g - V_g^0| \leq \delta_g$; exact equality may be infeasible with fixed atoms.
- Enforce connectivity, boundary links, neck size and protected cells separately.
- Bound refinement and patch expansion; report unresolved if no candidate passes.

The objective is proposed. A finite candidate search guarantees termination or rejection, not a feasible repair or a global optimum. Geometric subdivision must remain conforming across patch boundaries.



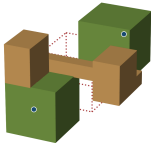
All voxel boundary links pass here, but
grain B has two volume components

The existing EBSD graph is a separate 2D workflow. Its pixel connectivity does not certify a 3D grain hull. Periodic volume components must be counted after face identification.

Three separate questions

1. Connected volume? Use BFS on shared faces.
2. Valid local surface? Check edges and vertex links.
3. Usable FE geometry? Check intersections, feature size and mesh quality.

A connected grain can still have a local self contact. Multiple boundary shells may also describe a valid cavity.



Adjacency follows shared cell faces

1. Locate the defective grain sectors.
2. Generate K candidate cell paths, ranked by changed volume and APD cost.
3. Expand paths to the required neck cross-section.
4. Branch over donor rerouting or another target-grain path.
5. Limit joint edits to D ; validate every complete candidate.

Contract exterior components to fixed terminals and preserve their connections.

Proposed candidate generator, not implemented. Paths do not certify topology. Limit search, patch growth and refinement; report unresolved if none passes.

Repository sources

- `examples/RVE_generation/create_rve.py`: adapted packing and voxel workflow.
- `tests/test_power_diagram_diagnostics.py`: controlled APD pinch.
- `tests/test_apd_boundary.py`: ordinary junction control.
- `apd_geometry.py`, `plotting.py`, `voxelization.py`: surfaces, images and voxel checks.
- `geometry_conditioning.py`, `surface_regularization.py`: existing baselines.

Every grain image uses Kanapy labels or Kanapy surface triangles. Source geometries, parameters and measurements are saved with the figure script.

The old PyVista example uses an obsolete global vertex key. Figures use the current APD data structures and Matplotlib. The contextual RVE is illustrative; no actual failing specimen from Hartmaier was available.

- Buze et al. (2024), Computational Materials Science 245, 113317: APDs and generalized Lloyd conditioning; empirical improvement is not a topology theorem.
- Zhang, Cha and Bajaj (2015), Procedia Engineering 124, 187–199: repair material classification before extracting shared surfaces.
- Hestroffer and Beyerlein (2022), XtalMesh: comparator for polycrystal volume meshing.
- CGAL Mesh_3 documentation: volume meshing options do not repair invalid input topology.

These sources support the architecture and provide comparators. They do not establish the proposed repair's robustness or preservation of microstructural statistics.